



Using fuzzy cognitive maps to develop policy strategies for the development of green rural housing: A case study in China

Xiaolong Gan^{a,*}, Kangkang Yan^b, Tao Wen^c

^a Center for Inclusive Finance and Agricultural & rural Development & School of Economic and Management, Southwest University, Chongqing 400715, China

^b School of Economic and Management, Southwest University, Chongqing 400715, China

^c Center for Inclusive Finance and Agricultural & rural Development & School of Economic and Management, Southwest University, Chongqing 400715, China

ARTICLE INFO

Keywords:

Green building
Rural housing
Fuzzy cognitive map
Policy mixes

ABSTRACT

Under the strategy of rural revitalization, China has a considerable demand for building green rural housing. Thus, it is urgent to develop related policy strategies for green rural housing development. However, the issue of green rural housing development has been overlooked in recent studies. This research fills this gap by increasing our understanding of factors that hinder green rural housing development and identifying effective policy strategies via fuzzy cognitive maps (FCM). Firstly, 23 system variables were identified including 8 policy drivers, 12 system structure variables and 3 system outcome variables. Then, the causal-effect relationships among these variables were visualized. Secondly, a fuzzy inference simulation was conducted to explore the most effective policy strategies for green rural housing development. The results show that these factors inhibiting green rural housing development are interrelated, which form solid vicious cycles not only between economic factors, social factors and technological factors but also within these sub-aspect factors. Three policy mixes were identified as the most effective policy strategies: financial support and price support, infrastructure development and technology support, guiding information and certification and labelling. The results offer a valuable reference to policy-making on facilitating rural green housing development.

1. Introduction

With the rapid economy and societal development, China has made a great achievement in the construction of rural housing. In 2019, the average rural living space is 47.3 m² per capita, which is 5.4 times that of 1979 (MHURD, 2020). Besides, rural housing has become safer and more risk-resistant due to shanty building revamp projects (Z.-T. Li et al., 2019). Despite this, due to the extensive adoption of rough construction methods and materials, the quality, functions and energy performance of rural housing cannot be guaranteed (Liu et al., 2014; Zhou et al., 2019). For instance, the comfort level is quite low due to the poor thermal environment and indoor air quality (Evans et al., 2014). Especially, rural residential energy consumption is nearly 924.6 billion kWh, which is the highest of all building types (He et al., 2014a). As the income grows, rural residential energy consumption is expected to reach 25 % of China's building energy use in 2050 (Evans et al., 2014). Consequently, rural residential quality has not been improved, and a large number of rural housing was rebuilt in a short time (Zhou et al., 2019). This leads to several negative environmental impacts on the

countryside as well as severe resource waste. Meanwhile, China has to confront a long-term threat regarding energy consumption as about 600 million live in rural areas (Evans et al., 2014; Shan et al., 2015).

Green building (GB) refers to the sustainable practices of applying resource-efficient and environmentally responsible processes and building structures throughout the life of a building (G. Wang et al., 2021). Lessons derived from the countries with good performance of GB development indicate that adopting GB technology can effectively address the problems in rural housing given its inherent benefits, including increased energy efficiency, less energy demand, reduced water usage, increased indoor quality and minimized construction waste (Nguyen et al., 2017). Currently, GB has been used by China's government as a tool to improve rural residential conditions. In 2009, the issued policy of "Notice on the construction of green rural houses" required conducting pilot works on green rural housing. Environmentally sensitive areas such as the Beijing-Tianjin region, the Yangtze River Delta and the Pearl River Delta should take the leading role in green rural housing construction. In 2019, a "Notice on the pilot work of rural housing construction" was issued, encouraging the adoption of local construction

* Corresponding author.

E-mail address: tomkan@swu.edu.cn (X. Gan).

<https://doi.org/10.1016/j.techfore.2023.122590>

Received 20 June 2022; Received in revised form 7 March 2023; Accepted 13 April 2023

Available online 28 April 2023

0040-1625/© 2023 Elsevier Inc. All rights reserved.

materials, and new construction methods. Meanwhile, the pilot and demonstration work of green rural housing construction was highlighted again. Overall, green rural housing construction stays in the pilot phase and has not been extensively applied in rural areas.

Rural housing in China is primarily self-built (Liu et al., 2014). Self-built housing refers to occupants who are involved in housing production for self-occupant (Bossuyt et al., 2018; Heffernan and Wilde, 2020). It is a cost-effective way for builders to access housing with better quality and more attuned to occupants' needs (Bossuyt et al., 2018). More importantly, self-built housing has been asserted to be more sustainable than housing built by developers. Housings built by developers are for sale to deliver a profit, and the risks of benefit-cost mismatch seriously hinder developers toward the adoption of green practices and technologies (Zhang et al., 2018). In contrast, self-built housing is built to live in, and decisions made by the builders regarding housing production are considered on a whole-life basis (Heffernan, 2015). Investing in enhanced energy performance becomes a sensible option. This view is supported by developed countries, such as the UK, which adopted the self-built method as a development model for sustainable homes (Heffernan and Wilde, 2020). Currently, a rural revitalization strategy has been issued by China's government to build a pleasant living environment, and it is estimated that about 5 billion housing units will be newly built or rebuilt in the next decades (Gan et al., 2022b). This provides an unprecedented opportunity for green rural housing development. Thus, promoting the development of green rural housing construction is urgent as self-built rural housing is suitable to develop green housing.

Facilitating green self-built housing development is essentially a transition of technology paradigm, requiring behavioural, technological, and institutional changes (Ornetzeder and Rohracher, 2003). It is impossible to complete the green transition of rural housing construction without promotion from the government, as innovations always encounter various interrelated barriers, especially in the initial development stage (Chang et al., 2019). Despite China's government having issued relatively systematic measures to stimulate the vitality of GB development, much attention to GB policy has been given to urban areas. Due to the different build modes, the policies and strategies for urban GB development are inapplicable to rural GB development. Therefore, developing policy strategies is the precondition for the development of green rural housing. As highlighted by Ladu et al. (2020), facilitating the development of green rural housing is rested on concerted policy efforts across a wide range of policy spheres. In other words, individual policy instrument is inadequate to induce technological transitions, which depend on the policy mix (Falcone et al., 2017). Policy formulation for technological transition lies in the combination of technology push instruments and technology demand pull instruments as well as relevant actors (Falcone et al., 2019). Involving local stakeholders in policy designing for technological transitions is crucial which can increase social acceptability as well as cooperation between local stakeholders (Falcone et al., 2017).

A few studies of green rural housing have been conducted including the aspects: 1) analysing the performance of green rural housing (Abanto et al., 2017; Garg et al., 2016); 2) examining the development status of green rural housing (He et al., 2014a; He et al., 2014b); 3) developing technical measures for green rural housing construction (Dong and Jin, 2013). These studies provide a solid foundation for green rural housing development, while limited studies have endeavoured to develop policy strategies by exploring the interrelated barriers to green rural housing development. As a result, there are some unsolved questions as follows:

- (1) What are the key barriers to green rural housing development in China?
- (2) Which drivers have the greatest potential in facilitating green rural housing development in China?

- (3) What policy actions might be more effective in facilitating green rural housing development in China?

This study employs the method of fuzzy cognitive maps (FCM) to answer the above questions. FCM is a participatory modelling method that determines the behavioural complexity of a system based on causal-effect relationships (Falcone and Rosa, 2020), which has been extensively used to formulate optimal policy strategies by understanding the inner system dynamics. It has been considered an effective tool for the aid decision maker in pursuit of sustainability. The final aim of this study is to propose suggestions for policy actions on the development of green rural housing. The results of this study contribute to improving the effectiveness and efficiency of policymaking for GB development by exploring the interactions between factors impeding GB development in China's rural housing. As the country with the largest agricultural population in the world, the research findings offer useful lessons for other developing countries.

This study contributes to the previous studies in the following aspects. First, this study focused on green housing development in rural areas which has been rarely explored in previous studies (Liu et al., 2020a; Saka et al., 2021; W. Wang et al., 2021; Yas and Jaafer, 2020). The implementation of rural revitalization strategy provides the best opportunity for green rural housing development in China, which needs to be given attention by developing policy strategies. Second, this study comprehensively identifies the factors that impeded the green housing development regarding rural self-built housing. To the authors' best knowledge, limited studies have explored the factors that impeded the green building development related to self-built housing as much attention have been given to housing developer in urban areas (Hwang and Wei, 2013; Li et al., 2014; Nelms et al., 2005; G. Wang et al., 2021; Zhao et al., 2020). Third, concerning previous studies using the method of FCM (Gan et al., 2019; Hong et al., 2013; Huang et al., 2013; Kontogianni et al., 2012; Kyriakarakos et al., 2014; Obiedat and Samarasinghe, 2016), this studies not only explored the interrelationships between the system variables but also identified the best-performed policy strategies in different scenarios. The results of this study provided valuable references for the policy-making of green rural housing development in different rural areas.

The rest of this research is organized as follows: a literature review regarding the factors impeding green rural housing development was provided in Section 2. Section 3 introduces the theoretical background of FCM. Section 4 describes the simulation process of Fuzzy cognition maps. Section 5 reports the results of this research. Section 6 discusses the results of this research. Section 7 provides the conclusions of this research as well as the limitations.

2. Literature review

2.1. Factors impeding the development of green rural housing

GB as a viable approach to achieving sustainable goals has been well acknowledged, while in many parts of the world, GB development is found to be insufficient (Chan et al., 2017). GB development is not only a technocratic exercise, but also a social-technical transition, involving changes within policies, social norms, technology, and so on (Deng et al., 2018). It is indeed a tough task as the such transition is facing a lot of factors that hinder the extensive adoption of GB practices and technologies. Many efforts were carried out on the factors impeding GB development in urban areas, while the development of green rural self-built housing has been rarely studied. A systematic method was used to search and review related literature with keywords of "barrier", "factors", "rural housing", "green building", "self-built" and other associated terms. Web of Science database was selected. Concerning the analysis framework extensively adopted in construction innovation (Chan et al., 2017; Liu et al., 2020b; Yas and Jaafer, 2020), factors impeding GB development in rural self-built housing could be divided into three

categories: economic, technological, social factors.

2.1.1. Economic factors

Economic factors are usually the highest priority when innovative technologies are introduced (Gan et al., 2015). In China, rural households have a per capita disposable annual income of USD 1772, far below that of urban households (Ren et al., 2019), whereas the initial investment for green projects ranges from 1 % to 25 % higher than conventional projects (Hwang and Wei, 2013). Rural green housing is often beyond the income capacity of most low-income earners, considering the incremental cost (Adabre et al., 2020). Thus, higher investment has been considered the topmost and undeniable factor impeding the adoption of GB practices and technologies for builders (Chan et al., 2017). Moreover, lenders are likely to regard self-built loans as having higher risk, the financial support is inaccessible for self-builders (Gan et al., 2022b; Wilson, 2017). It is deadly to GB development as builders would not be able to adopt GB practices and technologies without a better financial foundation (Chan et al., 2018). Finally, the primary GB materials and technologies are unavailable in rural areas (Chan et al., 2018).

2.1.2. Technological factors

Adopting GB technologies involves complex code and requirements, and self-builders always confront technical difficulties, as they are not skilled workers (Zhao et al., 2020). For instance, technical difficulties of GB adoption will be experienced during the construction process, and the design of green housing will be more complicated than that of conventional buildings (Hwang and Wei, 2013). More importantly, the lack of suitable GB technology and the absence of technical support is exceptionally outstanding in rural areas (Chan et al., 2018; Qin et al., 2016). For example, lots of mature cleaner energy technologies, such as photolytic (Solar) panels are unsuitable for rural housing because of high-cost investment (Gan et al., 2022b; Sullivan and Ward, 2012). At the same time, as Wu and Xu (2013) highlighted, the technical level of Chinese GB is relatively less developed than that of developed countries primarily concentrated in urban areas. The low technical level of Chinese domestic GB remains a challenge because of low investment in research and development (R&D) (Liu et al., 2020b). This leads to the limited number of novel green technologies for Chinese rural areas.

2.1.3. Social factors

Social factors highlight the awareness, beliefs, and attitudes of self-builders regarding GB. Darko and Chan (2016) pointed out that lacking information and knowledge has been considered the top barrier to the adoption of GB. This makes it difficult for rural households to understand the benefits of GB and increase their awareness of GB. Consequently, many people distrust the quality and performance of GB technologies (Chan et al., 2017), and persuading related stakeholders with inadequate knowledge and information about GB is impossible. Besides, rural housing construction usually follows traditional social norms on housing construction, which result in related stakeholders being deeply rooted in traditional construction methods, rejecting any change in their attitudes and behaviours concerning housing construction (Chan et al., 2018). Due to traditional social norms, they show little interest in GB practices and technologies when they are introduced. Unwilling to change traditional methods of housing construction poses a significant challenge to GB development in rural housing (Chan et al., 2017). Compounding their unwillingness for GB is a lack of demand for GB. The widespread perception is that the demand is lacking for green properties among the public (Zhang et al., 2011).

2.2. Policy driver for the development of green rural housing

Policies play a paramount role in facilitating the development of green buildings which has been well acknowledged globally, and the impacts of policies on facilitating the development of green buildings

have been verified. For instance, the green innovation policy is extremely critical for the development of green buildings in China's urban areas (Kong and He, 2021). The research of Zhang et al. (2018) indicates that land-related policies have significantly stimulated the appearance and development of green housing. With references to previous studies (Cui et al., 2019; Han et al., 2021; Y. Li et al., 2019; G. Wang et al., 2021; W. Wang et al., 2021; Yas and Jaafer, 2020), the policies on promoting the green rural housing development can be divided into demand-side and supply-side.

Demand-side policies include economic incentives, guiding information, price support, certification and labelling. Economic incentives bring financial gains for beneficiaries, such as financial subsidies, tax incentives, and preferential loans, which are monetary and commonly provided by governments (Ayokunle et al., 2016). It has been used to address the prominent barriers to green building development, e.g., high initial investment and long payback period. Guiding information aims to persuade rural residents to convert their behaviour by affording information and cases of successful implementation of green rural housing (Huang et al., 2016). For example, the US federal government are committing to adopting GB which aims to lead by example (Nelms et al., 2005). Owners are motivated by GB technology when there are champions to promote them. Certification and labelling have been regarded as a gratifying incentive to improve their image and reputation, which gives credence and validation to the greenness of GB (Saka et al., 2021), and provides an avenue for improving competitive advantages (Ayokunle et al., 2016). Besides, there is a strong linkage between green building certifications and self-identity (Darko et al., 2017).

The main goal of supply-side policies is to enlarge the supply capacity of green buildings in rural areas and influence industrial innovation by stimulating R&D activities (Gan et al., 2022c). Supply-side policies include education and training, infrastructure development, technological support, and public services. The tool of education and training is extremely important for cultivating skilled workers for green housing development (Liu et al., 2020b; Nguyen et al., 2017). This is critical for the development of green rural housing as skilled workers are generally scarce. Infrastructure development can provide the needed infrastructure, such as R&D labs, for the innovation of green buildings, and cultivate or strengthen the capacity of construction enterprises for green buildings (Gan et al., 2022c). Technology support aims to strengthen the R&D activities of green rural housing in the way of national science and technology projects and scientific programs, which can address the issue of technology shortage. Public services highlight streamlining the procedures of management activities of green building development, such as procedures for approvals, planning, and others (Gan et al., 2022a).

3. FCM

FCM was proposed to model the behavioural complexity of a system using fuzzy causal relationships (Özesmi and Özesmi, 2004). It generates a graph which consists of nodes representing the variables within a system and links standing for the causal-effect relations among variables (Gan et al., 2019; Magazzino and Falcone, 2022). FCM permits feedback loops, enabling us to capture the dynamics of the nonlinear system and obtain a potential solution to specific problems (Lopolito et al., 2020; Obiedat and Samarasinghe, 2016). It is a semi-quantitative method that model how certain system dynamic evolve (Morone et al., 2021), which has been regarded as a simplified way of mathematically modelling a complex system and is suitable to deal with case studies lacking statistical data (Ladu et al., 2020; Reckien, 2014). As a participatory model approach, FCM enables local stakeholders to assert their visions which helps the government to understand local priorities in policy making (Falcone and Rosa, 2020), and consequently, great public acceptance of policy implementation can be achieved. The method of FCM has been used in multiple studies regarding sustainability transition, e.g., Reckien (2014), and Gan et al. (2019). Based on the previous studies, the main

steps of using the FCM are listed as follows:

Step 1. FCM construction. Nodes are connected by arcs (e_{ij}) with different weights (w_{ij}). The arcs (e_{ij}) refer to causal flows from node C_i to C_j , and the weights are expressed as positive, negative and neutral causality and take values in the spaces $[-1, 1]$. A qualitative condensation approach is adopted to generate stakeholder-grouped FCM by aggregating individual FCM (Gan et al., 2019).

Step 2. FCM indices analysis. The FCMs for all the stakeholders can be transformed into a $[n \times n]$ weight matrix, gathering the weights w_{ij} of the interconnections between n nodes of the FCM. Based on the graph theory and social network analysis, the FCM indices include density, centrality, and centrality (Hong et al., 2013).

The density index can be measured by Eq. (1):

$$D = \frac{C}{N^2} \quad (1)$$

where N is the number of variables and C represents the number of interconnections. The centrality [Cen] is examined by Eq. (2):

$$Cen_D(c_i) = od(C_i) + id(C_i) \quad (2)$$

The in-degree $od(C_i)$ and out-degree $id(C_i)$ can be calculated by Eqs. (3) and (4)

$$od(C_i) = \sum_{j=1}^N \bar{w}_{ij} \quad (3)$$

where N is the number of nodes connected from the node C_i , and (w_{ij}) is the weight of the connection from the node C_i to node C_j .

$$id(c_i) = \sum_{j=1}^N \bar{w}_{ji} \quad (4)$$

where N is the number of nodes connected to the node C_i , and (w_{ij}) is the weight of the connection entering the node C_i from node C_j .

Step 3. FCM Simulation. The neural network method is used for system simulation. In each simulation, the value A_i of a variable C_i is calculated by calculating the effect of the interconnected variable c_j on the specific variable c_i . After a set of iterations, the concept values evolve until being stabilized at a fixed point. Then, a final vector A is obtained by Eq. (5).

$$A_i^{(k+1)} = f \left(A_i^k + \sum_{j=1}^N j \neq i A_j^k W_{ji} \right) \quad (5)$$

where $A_i^{(k+1)}$ is the value of the concept C_i at step $k+1$, $A_j^{(k)}$ is the value of the concept C_j at step k , f is the threshold function and W_{ji} is the weight of the interconnection between C_i and C_j .

FCM simulation enables us to analyze “what-if scenarios” to explore whether the policy scenario reaches desirable outcomes (Obiedat and Samarasinghe, 2016). The what-if scenarios can be defined by changing one or severe variables in the FCM, suggesting external policy interventions. Then, the initial steady status and the what-if scenarios can be compared to reveal the effect of different policy interventions. This provides valuable references to explore potential solutions to the research problems.

4. Research methodology

4.1. Identification of system variables

The identification of system variables was conducted by an in-depth literature review and followed by validation through experts. Firstly, an initial list of system variables was suggested based on the literature review. Then, an in-depth interview was conducted to validate the

rationale of the preliminary list of system variables. Each author of this study suggested a list of experts on green rural housing developments, and then, four experts from academic, industrial and governmental institutions who have rich knowledge of promoting green rural housing development were selected as the experts for in-depth interviews. The initial list of system variables was sent to these experts to check the rationality of these suggested concepts, and suggest additional system variables. The interviewees agreed with the identified variables, and the final list of system variables was obtained.

4.2. Building fuzzy cognitive maps

This step was aimed at building a fuzzy cognitive map based on the knowledge and perceptions of experts on green rural housing development. To this end, semi-structured interviews with critical stakeholders of the pilot GB project on rural housing in Chongqing, China was conducted. Four stakeholder categories, including rural residents, government officials, rural craftspeople, and managers of construction enterprises, were selected because they play a critical role in GB adoption in rural housing (Li et al., 2014; Zhou et al., 2019). First, an invitation to a semi-structured interview was sent to practitioners of pilot rural green housing by email or telephone, explaining the goal of a semi-structured interview. Then, 28 experts agreed to participate in the interviews, including 7 rural residents (25 %), 6 government officials (21.4 %), 7 rural craftspeople (25 %), and 8 construction enterprise managers (28.6 %).

The semi-structured interviews took place from June 24th to 30th 2021. During the interview, each participant was required to give their opinion on the strength of relations between paired system variables. The interrelationships between these factors were described using the linguistic term, such as very negatively high (−4), negatively high (−3), negatively medium (−2), negatively low (−1), zero (0), positively low (+1), positively medium (+2), positively high (+3) and very positively high (+4). Following this, the individual fuzzy cognitive map was built. Then, a qualitative condensation approach is adopted to generate stakeholder-grouped FCM by aggregating individual FCM, with the summed interrelationships converted and defused into numerical values in the interval $[-1, 1]$ (Gan et al., 2019). Finally, a computational model of FCM can be developed.

4.3. System analysis and simulation

System analysis was conducted through the method of social network analysis. Firstly, The FCMs for all the stakeholders can be transformed into a $[n \times n]$ weight matrix, aggregating the weights w_{ij} of the interrelationships between n concepts of the FCM. Secondly, some critical indices were used to assess the system configuration characterized by the interrelationships between variables, including network density, average degree, centrality degree (in-degree and out-degree), and clustering coefficient (Falcone and Rosa, 2020). The software of Ucinet 6.0 was employed to conduct the network analysis, and visualize the interrelationships between the system variables.

System simulation was conducted by using the method of artificial neural network (ANN), involving two steps. Firstly, the natural dynamic simulation is used to explore how the system evolves without external interventions, namely the initial state, and the most influential system variables can be determined by computing the system's steady-state values (Morone et al., 2021). Secondly, the what-if policy scenarios are adopted to simulate the effects of policy interventions, namely to explore what will happen if implementing related policy measures. Especially, changes from the initial steady-state value were examined after proceeding system simulation model under different what-if policy scenarios.

5. Results

5.1. Descriptive analysis

As shown in Table 1, a total of 23 system variables were identified from the literature review and in-depth interview, which can be categorized into policy driver, system structure variables and system outcome variables. Among these, the factors of sector structure can be divided into three sub-aspects: technological aspect, social aspect and economic aspect.

The logic that underlines the system variable grouping is presented in Fig. 1. Namely, policy drivers can alter system structure variables, producing direct and indirect effects on the system outcomes. The variables of policy drivers are senders, and experts are not permitted to suggest any incoming connections. System structure variables were transmitters, suggesting these variables can exert influence on other variables and be influenced by other variables. System outcomes variables were receivers, implying they were not allowed to indicate any out-coming connections.

5.2. Fuzzy cognitive mapping

The integrated fuzzy cognitive map was presented in Fig. 2, including 23 nodes and 206 direct links. Overall, system variables were quite interrelated based on the graph indexes as shown in Table 2. The average degree of the integrated fuzzy cognitive map suggests the average number of links per variable is 6.957. The results of cluster coefficient and average shortest path length further confirm the inter-related characterization of the integrated fuzzy cognitive map. The network appears to a relatively moderate as its 206 connections represent more than 40 % of the maximum theoretical number of direct connections among 23 nodes.

As presented in Table 3, among the ordinary nodes, the most influencing system structure variables with higher scores of out-degree, including lacking suitable technology (T2), lacking information and knowledge (S3) and lower income level (E2). While, the most influenced system structure variable with higher scores of in-degrees, including

Table 1
The identified system variable of FCM.

Dimension	Variable	Num		
Policy driver	Demand side	Guiding information	DD1	
		Financial support	DD2	
		Prices support	DD3	
		Certification and labelling	DD4	
	Supply Side	Education and training	DS1	
		Infrastructure development	DS2	
		Technological support	DS3	
		Public services	DS4	
	System structure	Economic aspect	Higher initial investment	E1
			Lower income level	E2
Unavailable product market			E3	
Unavailable financial support			E4	
Technological aspect		Technology difficulty in construction	T1	
		Lacking suitable technology	T2	
		Lacking skilled workers	T3	
		Uncertain performance of green rural housing	T4	
Social aspect		Lacking demand from rural residents	S1	
		Traditional social norms	S2	
		Lacking information and knowledge	S3	
		Skeptical attitude to green rural housing	S4	
System outcome		Reducing the life cycle cost	R1	
		Increasing living quality	R2	
	Improving self-identification	R3		

lacking information and knowledge (S3), lacking demand from rural residents (S1), and lacking suitable technology (T2). It deserves to notice that two factors, lacking suitable technology (T2) and lacking information and knowledge (S3), were perceived as strongly influencing and influenced variables simultaneously. The most influential policy drivers, based on the out-degree scores, were technological support (DS3), financial support (DD2), and guiding information (DD1). The most influential system outcome variables based on the in-degree scores were improving self-identification (R3), increasing living quality (R2), and reducing life cycle costs (R1).

By further examining the integrated fuzzy cognitive maps as shown in Fig. 2, it was found that policy drivers have a direct effect on the system's structural variables as well as outcomes variables. For instance, financial support (DD2) exert a direct influence on lacking demand from rural residents (S1), price support (DD3) has a direct influence on skeptical attitude to green rural housing (S4), education and training (DS1) can directly influence the lacking information and knowledge (S3). Meanwhile, price support (DD3) exerts a direct effect on reducing the life cycle cost (R1), certification and labelling (DD4) has a direct effect on improving living quality (R2), education and training (DS1) has a direct effect to the improving self-identification (R3). In the meanwhile, the indirect effect of the policy drivers on the system outcome variable can also be found. For instance, guide information (DD1) exerts influence on the system's structural variable, such as lacking information and knowledge (S3), traditional social norms (S2), and skeptical attitude to green rural housing (S4), and then, affects the system outcome variable, improving self-identification (R3). Infrastructure development (DS2) can exert influence on the unavailable product market (E3), higher initial investment (E1), and technological difficulties in construction (T1), and then, affect the reaching the life cycle cost (R1). Generally, the demand policy drivers affect the outcome variables by influencing the economic and social factors, while the supply policy driver influences the outcome variable by influencing the technological and social factors.

Related to the system structure variables, it was found that economic factors mainly affect social factors which are likely influenced by technological factors. Meanwhile, social factors have a direct influence on technological factors, and meanwhile, are influenced by economic factors. According to Gan et al. (2019), this suggests that the interactions among factors generate a vicious cycle impeding green rural housing development. The self-loops existed among the factors within sub-aspects. Affected by traditional social norms, rural residents usually lack information and knowledge of green housing, and consequently, they remain skeptical attitude on green housing. As a result, less demand for green housing was found in rural housing construction, which generally follows traditional social norms.

5.3. Fuzzy simulation

5.3.1. Natural dynamic simulation

To explore how the system evolves without interventions, scores of system variables' steady-state were calculated under the scenario of zero policy driver. As shown in Table 4, the most influential system structure variables in decreasing order were lacking information and knowledge (S3), lower income level (E2), and lacking suitable technology (T2). Of note, these three variables were regarded as key system variables, which have been confirmed by their out-degree scores. This suggests that these variables can exert a significant influence on system outcomes when certain policies are implemented. The outcome variables in decreasing order were, increasing living quality (R2), reducing the life cycle cost (R1), and improving self-identification (R3).

5.3.2. Effects of individual policy intervention stimulation

In the individual policy intervention stimulation, eight policy scenarios related to individual policy driver were introduced. In each policy scenario, except for the selected policy driver, all policy drivers were set

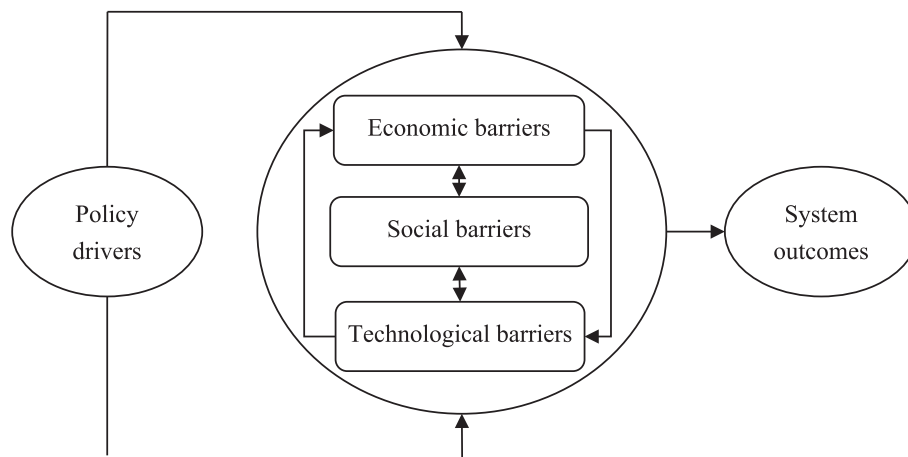


Fig. 1. The logic of system variables.

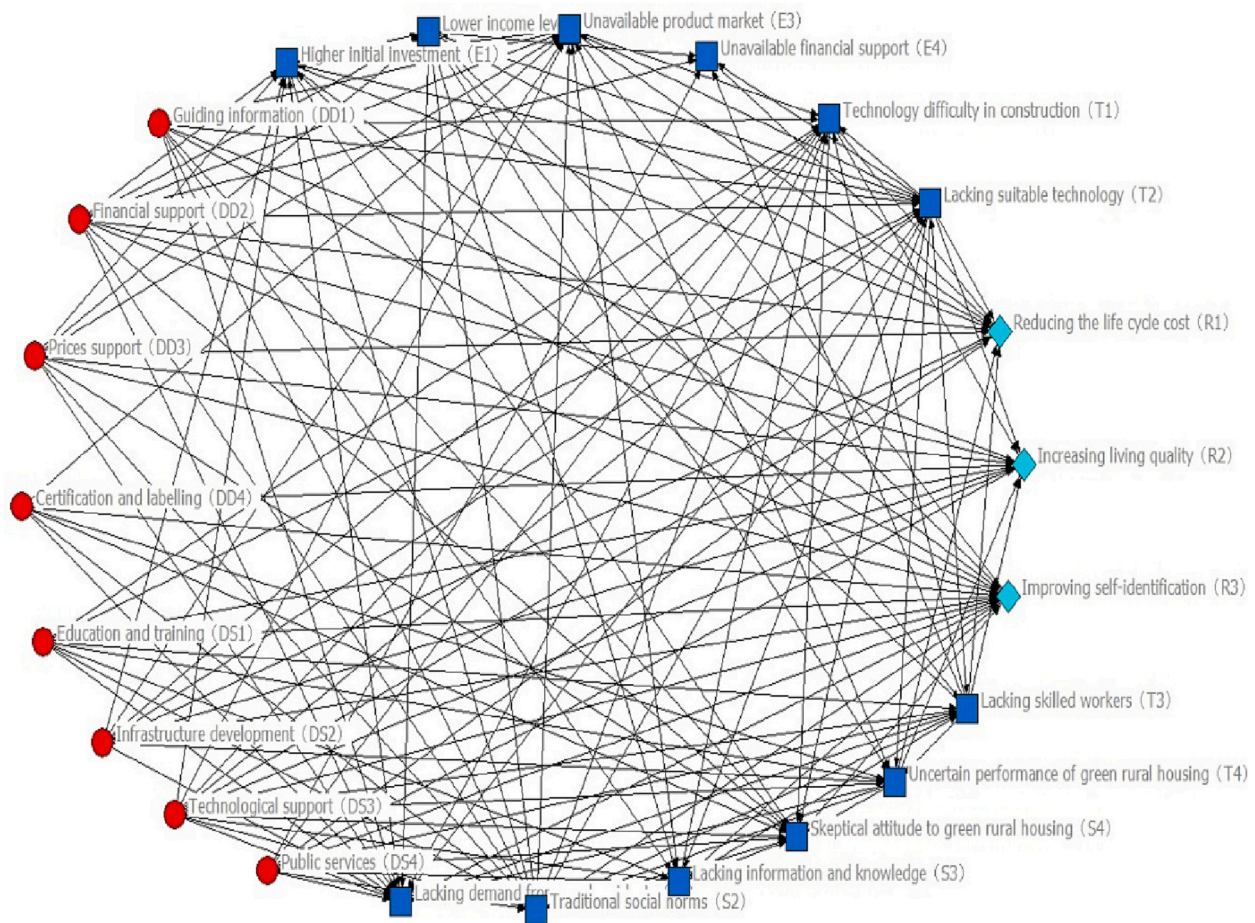


Fig. 2. Visualization of the integrated fuzzy cognitive map

to 0. With the help of the fuzzy inference algorithm mentioned in Section 3, variation from the zero policy scenario can be obtained, and the importance of the policy drivers was characterized in Fig. 3 to Fig. 5.

As displayed in Fig. 3, financial support (DD2) and price support (DD3) were found to be effective in all the economic factors. Of note, price support (DD3) showed the highest impact on the higher initial investment, and financial support (DD2) exerted the most significant impact on lower income levels and unavailable financial support. Public services (DS4) was the next best policy driver to unavailable financial support. Besides, infrastructure development (DS2), was thought to be

the most effective for the unavailable product market, followed by price support (DD3).

As shown in Fig. 4, supply-side policy drivers seem to be more effective in addressing technological factors. Among those, technological support (DS3) was found to be the most useful driver of three factors including technology difficulty in construction, lacking suitable technology and uncertain performance of green rural housing. This is followed by education and training (DS1) and infrastructure development (DD3). The driver of education and training (DS1) was the most effective driver of the lacking skilled workers.

Table 2
Graph index of fuzzy cognitive maps.

Graph index	The total
Number of links	206
Number of transmitter nodes	8
Number of receiver nodes	3
Number of ordinary nodes	12
Average degree	6.957
Average shortest path length	1.490
Clustering coefficient	0.344
Density	0.316

Table 3
Results of network analysis of integrated FCM.

Variable	Code	In-degree	Out-degree	Central degree
Guiding information	DD1	0.000	3.867	3.867
Financial support	DD2	0.000	4.476	4.476
Prices support	DD3	0.000	3.777	3.777
Certification and labelling	DD4	0.000	3.567	3.567
Education and training	DS1	0.000	3.566	3.566
Infrastructure development	DS2	0.000	2.834	2.834
Technological support	DS3	0.000	5.366	5.366
Public services	DS4	0.000	2.667	2.667
Higher initial investment	E1	3.399	2.400	5.799
Lower income level	E2	4.189	3.099	7.288
Unavailable product market	E3	3.834	1.901	5.734
Unavailable financial support	E4	1.100	0.766	1.866
Technology difficulty in construction	T1	3.355	1.633	4.988
Lacking suitable technology	T2	4.365	5.298	9.664
Lacking skilled workers	T3	1.467	2.100	3.566
Uncertain performance of green rural housing	T4	3.266	2.233	5.499
Lacking demand from rural residents	S1	4.543	1.300	5.843
Traditional social norms	S2	3.143	2.389	5.532
Lacking information and knowledge	S3	4.866	4.165	9.032
Skeptical attitude to green rural housing	S4	3.577	1.033	4.610
Reducing the life cycle cost	R1	4.933	0.000	4.933
Increasing living quality	R2	5.333	0.000	5.333
Improving self-identification	R3	7.066	0.000	7.066

Table 4
Simulation results of zero policy scenario.

Variable	Code	Steady-state value
Guiding information	DD1	0.000
Financial support	DD2	0.000
Prices support	DD3	0.000
Certification and labelling	DD4	0.000
Education and training	DS1	0.000
Infrastructure development	DS2	0.000
Technological support	DS3	0.000
Public services	DS4	0.000
Higher initial investment	E1	0.853
Lower income level	E2	0.957
Unavailable product market	E3	0.878
Unavailable financial support	E4	0.754
Technology difficulty in construction	T1	0.919
Lacking suitable technology	T2	0.953
Lacking skilled workers	T3	0.847
Uncertain performance of green rural housing	T4	0.912
Lacking demand from rural residents	S1	0.945
Traditional social norms	S2	0.934
Lacking information and knowledge	S3	0.959
Skeptical attitude to green rural housing	S4	0.942
Recondition the life cycle cost	R1	0.073
Increasing living quality	R2	0.099
Improving self-identification	R3	0.035

As denoted in Fig. 5, demand-side policy drivers were found to be more useful regarding social factors. Guiding information (DD1) was extremely useful in tackling the factors of traditional social norms, and lacking information and knowledge. Meanwhile, the driver of education and training (DS1) was also thought to be useful in tackling these factors. Financial support (DD2) showed the highest influence on the lacking demand from rural residents and followed by price support (DD3). Certification and labelling exerted the highest influence on the skeptical attitude to green housing. Guiding information (DD1) was also useful in addressing the issues of skeptical attitudes to green housing.

5.3.3. Effects of policy mixes stimulation

In this simulation, the influences of combined policy drivers were examined. Considering the scarcity of resources, each policy mix contains no more than two policy drivers as the simultaneous implementation of multiple policies in vast rural areas is neither achievable nor affordable for China's government (Morone et al., 2021). Therefore, the policy invention scenarios which combined two policy drivers are developed in Table 5. What we should bear in mind is that the value of variations does not highlight the magnitude of the variables, but is an index that permits speculation of the importance of the variables, which permits us to conduct the comparative analysis of the measurements (Falcone et al., 2017). Changes from the zero policy scenario simulation were calculated for each policy mix and were presented in Table 6.

Based on the ranking as shown in Table 6, policy mix 8 combining the financial support (DD2) and price support (DD3), showed the largest variations in the outcome variable of reducing the whole life cycle cost (R1). Regarding the increasing living quality (R2), the policy mix 26 combining the infrastructure development (DS2) and technology support (DS3), created the highest variations. Policy mix 3 combining guiding information (DD1), and certification and labelling (DD4), was expected to exert a significant influence on the outcome variable of improving self-identification.

6. Discussions

Housing is the core of human settlement, which is among the primary social conditions that define the life quality as well as citizens' well-being of any nation (Chan and Adabre, 2019). Producing sufficient housing in a socially and ecologically responsible manner plays a crucial role in achieving sustainable development at various levels, such as in cities, towns and countries (Gan et al., 2015). The long-term development priority given to cities leads to a significant gap between urban and rural areas regarding sustainable housing development (Wang et al., 2019). This gap is extremely acute in developing countries, such as China, where rural decline is undisputed. The proposed rural revitalization strategy in China stimulates rural development and aims at narrowing the gap between urban areas and rural areas. Thus, promoting the rural GB development to improve the rural settlement environment is necessary and urgent as billions of housing units will be newly developed or rebuilt (Zhou et al., 2019). China's rural housing is mainly self-built, which has been asserted to be suitable for adopting GB technologies. Despite this, the low uptake of GB practices and technologies by rural households suggests that there is a potential market failure of GB adoption in rural housing. Overcoming the market failure largely depends on the government to issue related policy strategies to facilitate green rural housing development. Considering the priority of GB development has been given to urban areas, China's government should pay more attention to policy-making on the development of green rural housing.

Facilitating the development of green rural housing is in essence a bottom-up method for the green transition of housing construction. This study explored the factors that impede GB development in rural housing which has been rarely studied in previous studies (Darko and Chan, 2016; Darko et al., 2017; Li et al., 2014; Liu et al., 2020b; Wu and Xu, 2013; Zhang et al., 2018; Zhang et al., 2011). Using the method of semi-

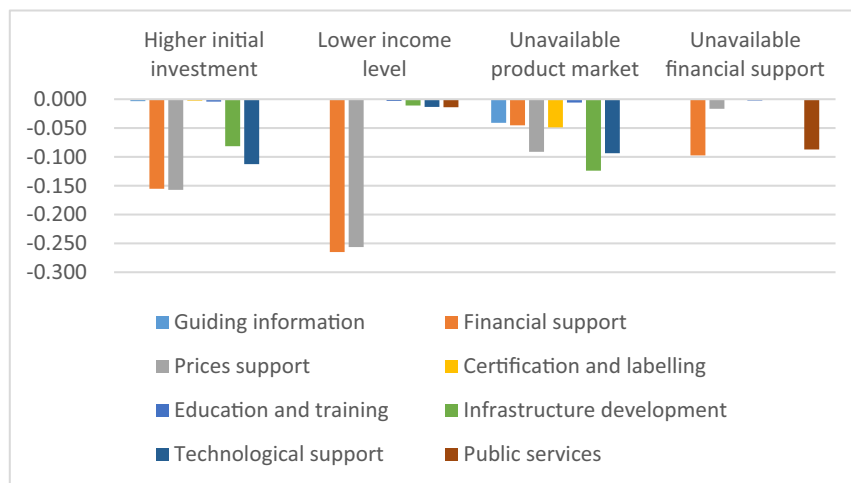


Fig. 3. Policy scenario simulation regarding economic factors.

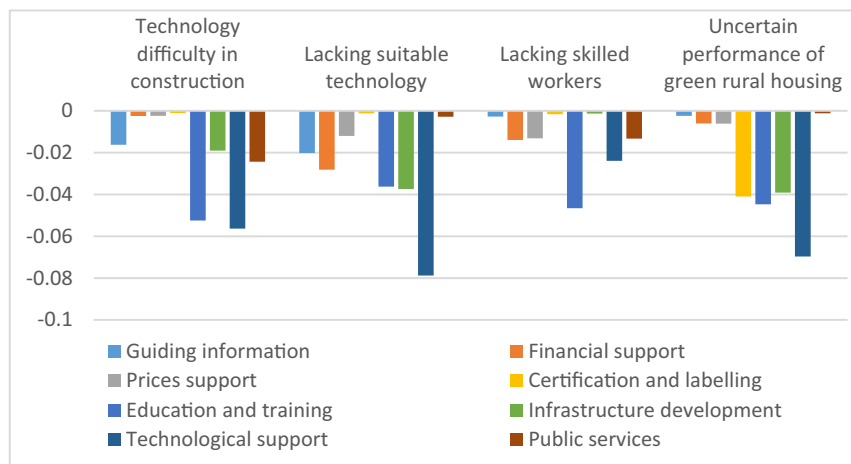


Fig. 4. Policy scenario simulation regarding technological factors.

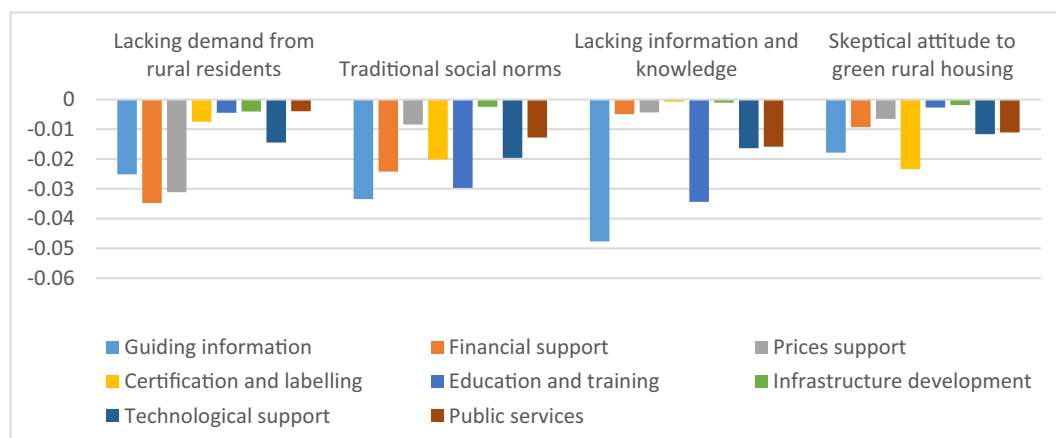


Fig. 5. Policy scenario simulation regarding social factors.

structured interviews, 12 factors impeding the GB development in rural housing have been identified, which can be divided into three categories, such as economic, social, and technological factors. It deserves to point out that, the issue of land supply and procurement has been highlighted as a critical factor for sustainable self-built housing

development in west developed countries (Wilson, 2017), which was not mentioned in this study. This is attributed to the different land systems. In west developed countries, land resources are dominated by market mechanisms; while in Chinese rural areas, all rural households are entitled to land for housing construction (Wang et al., 2012). Comparing

Table 5
Scenarios developing of policy mixes.

Policy drivers		DD1	DD2	DD3	DD4	DS1	DS2	DS3
DD1	Guiding information							
DD2	Financial support	Policy mix 1						
DD3	Prices support	Policy mix 2	Policy mix 8					
DD4	Certification and labelling	Policy mix 3	Policy mix 9	Policy mix 14				
DS1	Education and training	Policy mix 4	Policy mix 10	Policy mix 15	Policy mix 19			
DS2	Infrastructure development	Policy mix 5	Policy mix 11	Policy mix 16	Policy mix 20	Policy mix 23		
DS3	Technological support	Policy mix 6	Policy mix 12	Policy mix 17	Policy mix 21	Policy mix 24	Policy mix 26	
DS4	Public services	Policy mix 7	Policy mix 13	Policy mix 18	Policy mix 22	Policy mix 25	Policy mix 27	Policy mix 28

Table 6
Scenarios effect of alternative policy mixes.

Ranking	R1: Reducing the life cycle cost		R2: Improving living satisfaction		R3: Improving self-identification	
1	0.230	mix8	0.237	mix26	0.272	mix3
2	0.211	mix12	0.224	mix24	0.252	mix6
3	0.202	mix17	0.217	mix28	0.250	mix21
4	0.180	mix11	0.204	mix21	0.238	mix7
5	0.171	mix16	0.187	mix17	0.235	mix22
6	0.157	mix26	0.187	mix19	0.231	mix4
7	0.157	mix13	0.184	mix6	0.228	mix19
8	0.154	mix10	0.181	mix22	0.213	mix28
9	0.146	mix18	0.180	mix12	0.209	mix24
10	0.144	mix15	0.167	mix25	0.205	mix1
11	0.134	mix1	0.163	mix20	0.202	mix9
12	0.129	mix9	0.149	mix14	0.195	mix2
13	0.128	mix24	0.146	mix3	0.193	mix25
14	0.128	mix28	0.145	mix23	0.193	mix14
15	0.124	Mix2	0.142	mix9	0.191	mix5
16	0.119	mix14	0.138	mix27	0.190	mix20
17	0.105	mix6	0.131	mix15	0.183	mix12
18	0.100	mix21	0.129	mix4	0.174	mix17
19	0.098	mix27	0.125	mix18	0.173	mix26
20	0.096	mix23	0.123	mix10	0.164	mix13
21	0.078	mix25	0.123	mix7	0.158	mix10
22	0.076	mix5	0.117	mix13	0.154	mix18
23	0.072	mix20	0.106	mix16	0.150	mix27
24	0.058	mix7	0.101	mix5	0.148	mix15
25	0.055	mix4	0.097	mix11	0.144	mix23
26	0.052	mix22	0.087	mix2	0.124	mix8
27	0.050	mix19	0.084	mix8	0.119	mix11
28	0.032	mix3	0.079	mix1	0.110	mix16

Notes: the numeric value refers to the variation from the zero policy scenario, the policy mix refers to different policy scenarios, green indicates the highest values, and red suggests the lowest values.

private land ownership, collective land ownership has potential cost advantages for GB development as the less initial investment is required.

Via FCM, it was found that varied factors play different roles in impeding GB development in rural housing. Factors with a higher centrality degree include lacking suitable technology (T2), lacking information and knowledge (S3) and lower income level (E2). This suggests that addressing these issues is highlighted by stakeholders in moving the development of green rural housing. However, some critical factors, such as high initial investment (E1), mentioned in other studies (Chan et al., 2018; Darko and Chan, 2016; Deng et al., 2018), were not regarded as the most central factors. This difference can be explained that central factors are more interacted with other factors, and play an important role in the causal flow of the causal network (Zare et al., 2021). While the critical factors in other studies are decided by direct

perception. Complex causalities often exist in innovation stemming from multiple areas and are difficult to determine by direct observation. The method of FCM captures the complexity of the interactions between these factors that affect the examined systems based on a participatory approach (Falcone and Rosa, 2020). This method provides an innovative perspective to analyze GB development by illustrating the complex interactions between factors and can improve strategy performance.

The complex interrelationships between these factors have been presented in Fig. 2. As mentioned in Section 5.2, the complicated mutual interactions between economic, social and technological factors generate a vicious cycle, blocking GB development in rural housing. Originally, self-builders were given the power to make decisions on housing construction, and they are likely to select building strategies based on maximizing utility (Bossuyt, 2021). However, only if self-

builders have a sound understanding of the benefits of GB can they invest in improving housing energy performance (Heffernan and Wilde, 2020). Moreover, adopting sustainable technologies is often beyond the income capacity of most rural residents without government inventions (Adabre et al., 2020), and technological barriers make adopting green technologies infeasible for most rural self-builders (Sullivan and Ward, 2012). Moreover, the interrelationships between the factors within sub-aspects, as mentioned in Section 5.2, further reinforced the vicious cycle blocking the application of GB in rural housing construction. This makes the challenges in delivering sustainable housing too embedded, culturally and institutionally, to be merely framed as succumbing to several barriers. Consequently, despite China's government has encouraged the GB adoption in rural areas, rural residents often resist the sustainable mode of housing production due to these existed vicious cycles.

Another interesting finding as shown in Table 4 implies that, without external perturbations, the system would not evolve toward system outcomes, such as "reduction of whole life cost", "increasing living quality" and "improving farmer self-identification". This suggests that rural housing would persist with the traditional construction model without external interventions. In other words, the interrelated system structure variables forming a vicious cycle impeded the development of green rural housing. Meanwhile, the top three structure variables as presented in Table 4 are mainly concentrated on social aspects, lacking demand from rural residents (S1), traditional social norms (S2), and skeptical attitude to green housing (S3). According to Ladu et al. (2020), this implies that increasing social awareness of rural residents is recognized by stakeholders as a critical precondition to pushing the development of green rural housing. This is in line with previous studies highlighting the extremely critical role of social awareness in adopting GB technologies as behaviour is largely influenced by attitude, and subjective norms (Li et al., 2018; Ofek and Portnov, 2019; Portnov et al., 2018). According to Table 3, social factors not only exert influence on other aspect factors but also are largely influenced by other factors. Other factors have been found to have similar phenomena, such as economic factors and technological factors. This suggests that addressing the impediment factors to attracting rural residents to adopt green technologies in housing construction should be done via various policy measures.

A crucial finding from this study suggests that better development of green rural housing can be achieved through policy mixes compared with single policy measures. This indicates that policy mixes are usually sensitive to the interrelations between factors, and policymakers should take full account of this feedback and rebound effects into design suitable policy interventions. Meanwhile, as shown in Table 6, the best-performed policy mixes varied significantly based on the specific system outcomes. For instance, financial support (DD2) and price support (DD3) were perceived as key policy measures for reducing whole life cycle cost which was mainly influenced by economic and technological factors. Education and training (DS1), and technology support (DS3) were thought to be particularly useful in increasing living quality which is mainly affected by technological factors. Guiding and information (DD1) and certification and labelling (DD4) played a critical role in increasing living quality which was mainly affected by social factors. Different system outcome variables are affected by different system structure variables. Policy design on the development of green rural housing should avoid a simple one-fits-all approach, but dispose of the synergistic effect of the policy driver's interactions to facilitate the development of green rural housing.

7. Conclusions

This study explored and recommended the most effective policy strategies for green rural housing development via the method of FCM. A general fuzzy cognitive map was developed based on expert feedback, including 12 system structure variables, 8 policy drivers and 3 system outcome variables. The results of network analysis revealed that the

most significant influencing system structure variables are lacking suitable technology, lacking information and knowledge, and lower income level. The complex interrelationships were deduced and visualized, and the external and internal vicious cycles that block the development of green rural housing were found. The results of fuzzy stimulation indicate that promoting the development of green rural housing without policy intervention is extremely difficult. All the policy drivers exert a positive influence to some extent in facilitating the development of green rural housing individually. Three best-performed policy mixes have been found regarding different system outcome variables, such as financial support and price support, infrastructure development and technology support, guiding information and certification and labelling.

This study has some theoretical and practical contributions. Theoretically, this study offers the first application of the method of FCM to green rural housing development, to develop policy strategies to promote green rural housing development by understanding the interrelationships among these factors, which have been rarely explored in previous studies. Meanwhile, visualizing the interrelationships between these factors suggests the external and internal vicious cycles between and within the sub-aspect system structure impeding the development of green rural housing autonomously. Moreover, the interrelationships between these factors should be taken into due consideration in policy design as the existing feedback and rebound effect. Practically, this study suggests that the method of FCM is an effective participatory approach to aid policy design based on stakeholder involvement and a clear set of foreseen outcomes. It provides a clear departure for policy-making for the development of green rural housing. Secondly, the effects of policy mixes vary significantly according to different system outcomes. Policy design for the development of green rural housing should avoid the approach of one-fits-all to design well-suited policy strategies. Thirdly, this study provided the best-performed policy mixes based on rigorous FCM simulation. Indeed, the results suggested that the policy mixes combined with financial support, price support, guiding and information, certification and labelling, infrastructure development, and technology support for the development of green rural housing have a better performance in promoting the green rural housing development.

This study has several limitations. Firstly, the method of FCM is a qualitative method suggesting it can be used to explore the best policy mixes but not to measure each policy measure. In other words, the performance of policy drivers can only be compared instead of measured. Secondly, the macroeconomic or broader socio-political environment variables have not been considered in this study which is mainly researched from the perspective of rural residents. This can be researched in future by including all the related variables in the model. Thirdly, future empirical studies can be conducted in different places to obtain a fully comprehensive analysis of all possible development strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

No data was used for the research described in the article.

Acknowledgement

This study is supported by the Science Foundation of Ministry of

Education of China (20YJC630021), Chongqing Social Science Foundation (2020YBGL82, 2021NDQN62), Southwest University under the Fundamental Research Funds for the Central Universities (SWU2109209), Innovation Research 2035 Pilot Plan of Southwest University (SWUPilotPlan026), National Natural Science Foundation of China (71773099), National Social Science Foundation (21ZDA062, 20AZD080).

References

- Abanto, G.A., Karkri, M., Lefebvre, G., Horn, M., Solis, J.L., Gómez, M., 2017. Thermal properties of adobe employed in Peruvian rural areas: experimental results and numerical simulation of a traditional bio-composite material. *Case Stud. Constr. Mater.* 6, 177–191.
- Adabre, M.A., Chana, A.C., Darko, A., Osei-Kyeib, R., Abidoye, R., Adjei-Kumid, T., 2020. Critical barriers to sustainability attainment in affordable housing: international construction professionals' perspective. *J. Clean. Prod.* 253, 119995.
- Ayokunle, O.O., Bo, X.P., Martin, S., 2016. Green building incentives: a review. *Renew. Sustain. Energy Rev.* 59, 1611–1621.
- Bossuyt, D.M., 2021. The value of self-build: understanding the aspirations and strategies of owner-builders in the Homeruskwartier, Almere. *Hous. Stud.* 36 (5), 1–18.
- Bossuyt, D., Salet, W., Majoor, S., 2018. Commissioning as cornerstone of self-build housing. Assessing the constraints and opportunities of self-build in the Netherlands. *Land Use Policy* 77.
- Chan, A.P.C., Adabre, M.A., 2019. Bridging the gap between sustainable housing and affordable housing: the required critical success criteria (CSC). *Build. Environ.* 151, 112–125.
- Chan, A.P.C., Darko, A., Ameyaw, E.E., Owusu-Manu, D.G., 2017. Barriers affecting the adoption of green building technologies. *J. Manag. Eng.* 3 (3), 04016057.
- Chan, A., Darko, A., Olanipekun, A.O., Ameyaw, E.E., 2018. Critical barriers to green building technologies adoption in developing countries: the case of Ghana. *J. Clean. Prod.* 172, 1067–1079.
- Chang, R., Cao, Y., Lu, Y., Shabunko, V., 2019. Should BIPV technologies be empowered by innovation policy mix to facilitate energy transitions? - revealing stakeholders' different perspectives using Q methodology. *Energy Policy* 129, 307–318.
- Cui, L., Chan, H.K., Zhou, Y., Da, I.J., Lim, J.J., 2019. Exploring critical factors of green business failure based on Grey-Decision Making Trial and Evaluation Laboratory (DEMATEL). *J. Bus. Res.* 98, 450–461.
- Darko, A., Chan, A., 2016. Review of barriers to green building adoption. *Sustain. Dev.* 25 (3), 167–179.
- Darko, A., Zhang, C., Chan, A.P.C., 2017. Drivers for green building : a review of empirical studies. *Habitat Int.* 60, 34–49.
- Deng, W., Yang, T., Tang, L., Tang, Y.T., 2018. Barriers and policy recommendations for developing green buildings from local government perspective: a case study of Ningbo, China. *Intell. Build. Int.* 10 (2), 61–77.
- Dong, J., Jin, H., 2013. The design strategy of green rural housing of tibetan areas in Yunnan, China. *Renew. Energy* 49, 63–67.
- Evans, M., Sha, Y., Bo, S., Deng, Q., Jing, L., Delgado, A., 2014. Building energy efficiency in rural China. *Energy Policy* 64, 243–251.
- Falcone, P.M., Rosa, S.D., 2020. Use of fuzzy cognitive maps to develop policy strategies for the optimization of municipal waste management: a case study of the land of fires (Italy). *Land Use Policy* 96, 104680.
- Falcone, P.M., Lopolito, A., Sica, E., 2017. Policy mixes towards sustainability transition in the Italian biofuel sector: dealing with alternative crisis scenarios. *Energy Res. Soc. Sci.* 33, 105–114.
- Falcone, P.M., Lopolito, A., Sica, E., 2019. Instrument mix for energy transition: a method for policy formulation. *Technol. Forecast. Soc. Chang.* 148, 119706.
- Gan, X., Zuo, J., Ye, K., Skitmore, M., Xiong, B., 2015. Why sustainable construction? Why not? An owner's perspective. *Habitat Int.* 47, 61–68.
- Gan, X.L., Chang, R.D., Langston, C., Wen, T., 2019. Exploring the interactions among factors impeding the diffusion of prefabricated building technologies: fuzzy cognitive maps. *Eng. Constr. Archit. Manag.* 26 (3), 535–553.
- Gan, X., Liu, L., Wen, T., 2022. Evaluation of policies on the development of prefabricated construction in China: an importance-performance analysis. *J. Green Build.* 17 (1), 149–168.
- Gan, X., Liu, L., Wen, T., Webber, R., 2022. Modelling interrelationships between barriers to adopting green building technologies in China's rural housing via grey-DEMATEL. *Technol. Soc.* 70, 102042.
- Gan, X., Xie, K., Liu, H., Rameezdeen, R., Wen, T., 2022. A bibliometric and content analysis of policy instruments on facilitating the development of prefabricated construction in China. *Eng. Constr. Archit. Manag.* <https://doi.org/10.1108/ECAM-01-2022-0044>.
- Garg, V., Kotharkar, R., Sathaye, J., Rallapalli, H., Kulkarni, N., Reddy, N., Sarkar, A., 2016. Assessment of the impact of cool roofs in rural buildings in India. *Energy Build.* 114 (Feb.), 156–163.
- Han, S., Yao, R., Li, N., 2021. The development of energy conservation policy of buildings in China: a comprehensive review and analysis. *J. Build. Eng.* 38, 102229.
- He, B.J., Li, Y., Miao, Y., Mou, B., Zhou, Y., 2014. Overview of rural building energy efficiency in China. *Energy Policy* 69, 385–396.
- He, B.J., Yang, L., Ye, M., 2014. Building energy efficiency in China rural areas: situation, drawbacks, challenges, corresponding measures and policies. *Sustain. Cities Soc.* 11, 7–15.
- Heffernan, E.E., 2015. Delivering Zero Carbon Homes And Sustainable Communities: The Potential of Group Self-build Housing in England. University of Plymouth.
- Heffernan, E., Wilde, P.d., 2020. Group self-build housing: a bottom-up approach to environmentally and socially sustainable housing. *J. Clean. Prod.* 243, 118657.
- Hong, Z., Jie, S., Chao, S., Mengying, H., 2013. Human attitudes in environmental management: Fuzzy Cognitive Maps and policy option simulations analysis for a coal-mine ecosystem in China. *J. Environ. Manag.* 115, 227–234.
- Huang, S.C., Lo, S.L., Lin, Y.C., 2013. Application of a fuzzy cognitive map based on a structural equation model for the identification of limitations to the development of wind power. *Energy Policy* 63, 851–861.
- Huang, B., Mauerhofer, V., Geng, Y., 2016. Analysis of existing building energy saving policies in Japan and China. *J. Clean. Prod.* 112, 1510–1518.
- Hwang, B.G., Wei, J.N., 2013. Project management knowledge and skills for green construction: overcoming challenges. *Int. J. Proj. Manag.* 31 (2), 272–284.
- Kong, F., He, L., 2021. Impacts of supply-sided and demand-sided policies on innovation in green building technologies: a case study of China. *J. Clean. Prod.* 294, 126279.
- Kontogianni, A.D., Papageorgiou, E.L., Tourkolias, C., 2012. How do you perceive environmental change? Fuzzy Cognitive Mapping informing stakeholder analysis for environmental policy making and non-market valuation. *Appl. Soft Comput.* 12, 3725–3735.
- Kyriakarakos, G., Patlitzianas, K., Da Masiotis, M., Papastefanakis, D., 2014. A fuzzy cognitive maps decision support system for renewables local planning. *Renew. Sustain. Energy Rev.* 39, 209–222.
- Ladu, L., Imbert, E., Quitzow, R., Morone, P., 2020. The role of the policy mix in the transition toward a circular forest bioeconomy. *Forest Policy Econ.* 110, 101937.
- Li, Y., Yang, L., He, B., Zhao, D., 2014. Green building in China: needs great promotion. *Sustain. Cities Soc.* 11, 1–6.
- Li, Q., Long, R., Hong, C., 2018. Differences and influencing factors for Chinese urban resident willingness to pay for green housings: evidence from five first-tier cities in China. *Appl. Energy* 229, 299–313.
- Li, Y., Song, H., Sang, P., Chen, P.H., Liu, X., 2019. Review of Critical Success Factors (CSFs) for green building projects. *Build. Environ.* 158, 182–191.
- Li, Z.-T., Yuan, M.-J., Hu, M.-M., Wang, Y.-F., Xia, B.-C., 2019. Evaluation of ecological security and influencing factors analysis based on robustness analysis and the BP-DEMATEL model: a case study of the Pearl River Delta urban agglomeration. *Ecol. Indic.* 101, 595–602.
- Liu, W., Spaargaren, G., Mol, A.P.J., Heerink, N., Wang, C., 2014. Low carbon rural housing provision in China: participation and decision making. *J. Rural. Stud.* 35, 80–90.
- Liu, G., Li, X., Tan, Y., Zhang, G., 2020. Building green retrofit in China: policies, barriers and recommendations. *Energy Policy* 139, 111356.
- Liu, G., Tan, Y., Li, X., 2020. China's policies of building green retrofit: a state-of-the-art overview. *Build. Environ.* 169, 106554.
- Lopolito, A., Falcone, P.M., Morone, P., Sica, E., 2020. A combined method to model policy interventions for local communities based on people knowledge. *MethodsX* 7, 10087.
- Magazzino, C., Falcone, P.M., 2022. Assessing the relationship among waste generation, wealth, and GHG emissions in Switzerland: some policy proposals for the optimization of the municipal solid waste in a circular economy perspective. *J. Clean. Prod.* 351, 131555.
- MHURD, 2020. The Latest Standard of Per Capita Housing Area.
- Morone, P., Yilan, G., Imbert, E., 2021. Using fuzzy cognitive maps to identify better policy strategies to valorize organic waste flows: an Italian case study. *J. Clean. Prod.* 319, 128722.
- Nelms, C., Russell, A.D., Lence, B.J., 2005. Assessing the performance of sustainable technologies for building projects. *Can. J. Civ. Eng.* 32 (1), 114–128.
- Nguyen, H.T., Skitmore, M., Gray, M., Zhang, X., Olanipekun, A.O., 2017. Will green building development take off? An exploratory study of barriers to green building in Vietnam. *Resour. Conserv. Recycl.* 127, 8–20.
- Obiedat, M., Samarasinghe, S., 2016. A novel semi-quantitative Fuzzy Cognitive Map model for complex systems for addressing challenging participatory real life problems. *Appl. Soft Comput.* 48, 91–110.
- Ofek, S., Portnov, B.A., 2019. Differential effect of knowledge on stakeholders' willingness to pay green building price premium: implications for cleaner production. *J. Clean. Prod.* 251, 119575.
- Ornetzeder, M., Rohrer, H., 2003. User-led innovations, participation processes and the use of energy technologies. *Energy Policy* 34, 138–150.
- Özesmi, U., Özesmi, S.L., 2004. Ecological models based on people's knowledge: a multi-step fuzzy cognitive mapping approach. *Ecol. Model.* 176, 43–64.
- Portnov, B.A., Trop, T., Svechikina, A., Ofek, S., Akron, S., Ghermandi, A., 2018. Factors affecting homebuyers' willingness to pay green building price premium: evidence from a nationwide survey in Israel. *Build. Environ.* 137, 280–291.
- Qin, X., Mo, Y., Jing, L., 2016. Risk perceptions of the life-cycle of green buildings in China. *J. Clean. Prod.* 126, 148–158.
- Reckien, D., 2014. Weather extremes and street life in India—implications of Fuzzy Cognitive Mapping as a new tool for semi-quantitative impact assessment and ranking of adaptation measures. *Glob. Environ. Chang.: Hum. Policy Dimens.* 26, 1–13.
- Ren, H., Yuan, N., Hu, H., 2019. Housing quality and its determinants in rural China: a structural equation model analysis. *J. Housing Built Environ.* 34 (1), 313–329.
- Saka, N., Olanipekun, A.O., Omotayo, T., 2021. Reward and compensation incentives for enhancing green building construction. *Environ. Sustain. Indic.* 11 (9), 100138.
- Shan, M., Wang, P., Li, J., Yue, G., Yang, X., 2015. Energy and environment in Chinese rural buildings: situations, challenges, and intervention strategies. *Build. Environ.* 91, 271–282.

- Sullivan, E., Ward, P.M., 2012. Sustainable housing applications and policies for low-income self-build and housing rehab. *Habitat Int.* 36 (2), 312–323.
- Wang, H., Fubing, S.U., Wang, L., Tao, R., 2012. Rural housing consumption and social stratification in transitional China: evidence from a national survey. *Hous. Stud.* 27 (5), 667–684.
- Wang, Y., Zhu, Y., Yu, M., 2019. Evaluation and determinants of satisfaction with rural liability in China's less-developed eastern areas: a case study of Xinju. *Ecol. Indic.* 104, 711–722.
- Wang, G., Li, Y., Zuo, J., Hu, W., Nie, Q., Lei, H., 2021. Who drives green innovations? Characteristics and policy implications for green building collaborative innovation networks in China. *Renew. Sustain. Energy Rev.* 143, 110875.
- Wang, W., Tian, Z., Xi, W., Tan, Y., Deng, Y., 2021. The influencing factors of China's green building development: an analysis using RBF-WINGS method. *Build. Environ.* 188, 107425.
- Wilson, W., 2017. Self-build and custom build housing (England). Retrieved from. www.parliament.uk/commons-library.
- Wu, Y., Xu, R., 2013. Green building development in China-based on heat pump demonstration projects. *Renew. Energy* 53, 211–219.
- Yas, Z., Jaafer, K., 2020. Factors influencing green building projects spread in the UAE. *J. Build.Eng.* 27, 100894.
- Zare, S.G., Alipour, M., Hafezi, M., Stewart, R., Rahman, A., 2021. Examining wind energy deployment pathways in complex macroeconomic and political settings using a fuzzy cognitive map-based method. *Energy*. <https://doi.org/10.1016/j.energy.2021.121673>.
- Zhang, X., Platten, A., Shen, L., 2011. Green property development practice in China: costs and barriers. *Build. Environ.* 46 (11), 2153–2160.
- Zhang, L., Wu, J., Liu, H., 2018. Policies to enhance the drivers of green housing development in China. *Energy Policy* 121, 225–235.
- Zhao, X., Hwang, B.G., Jane, L., 2020. Job satisfaction of project managers in green construction projects: constituents, barriers, and improvement strategies. *J. Clean. Prod.* 246, 118968.
- Zhou, J., He, P., Qin, Y., Ren, D., 2019. A selection model based on SWOT analysis for determining a suitable strategy of prefabrication implementation in rural areas. *Sustain. Cities Soc.* 50, 101715.
- Xiaolong Gan** received doctor degree in Management Science and Engineering from Chongqing University, China, in 2014. Currently, he is an associate professor in the college of economy and management, Southwest University. His research interests include technology innovation, sustainable construction, and stakeholder management. He has published more than 30 academic articles, including 2 high-index papers, and meanwhile, hosted more than 10 social and natural projects supported by the National Natural Science Foundation of China, National Social Science Foundation, Science Foundation of the Ministry of Education of China, Chongqing Social Science Foundation, and so on.
- Kangkang Yan** received the B.S. degree in Business Administration from Southwest University, Chongqing, China, in 2021. He is currently working toward the M.S. degree in Business Administration with the school of Southwest University, Chongqing, China. His research interests include Green Farm Housing, Green Buildings and Farm Housing Policy Tools.
- Tao Wen** received doctor degree in Agricultural and forestry economic management from southwest University, Chongqing, China, in 2004. He is currently a professor in the college of economy and management, Southwest University. His main research interests include rural development. He has published more than 100 academic articles and hosted more than 10 national founded projects.